

Hashcat — Practical Tool Guide

Hash modes, attack modes, wordlists, rules, masks, sessions, --show, and the key rule: do not guess the mode.

➤ Abdullah Nasser Almsan — Personal Pentest Reference

Usage note: Hashcat is an offline tool. Examples are for labs and data you are explicitly authorized to process.

TYPE

Hash type

MODE

-m

ATTACK

-a

SHOW

Result

Contents

1. Basic syntax
2. Attack modes that matter
3. Finding the Hash Mode
4. Masks and Rules
5. Sessions and results
6. Workflow

1. Basic syntax

Hashcat depends on both Hash Mode (-m) and Attack Mode (-a).

KALI TERMINAL — Help / modes

```
hashcat -h
hashcat -hh | less
```

KALI TERMINAL — Wordlist mode

```
hashcat -m [MODE] -a 0 [HASH_FILE] [WORDLIST]
```

KALI TERMINAL — Show cracked entries

```
hashcat -m [MODE] [HASH_FILE] --show
```

Note: The most important rule: do not guess -m. Check the help/example hashes and the source of the hash.

2. Attack modes that matter

Mode	Meaning	When
-a 0	Straight / wordlist	Usually the first lab choice.
-a 1	Combination	Combine two lists.
-a 3	Mask / brute-force	When the password shape is known.
-a 6	Hybrid wordlist + mask	Word plus a patterned suffix.
-a 7	Hybrid mask + wordlist	Patterned prefix plus word.

3. Finding the Hash Mode

KALI TERMINAL — Search help

```
hashcat -hh | grep -i "NTLM\bCRYPT\|SHA2-512"
```

KALI TERMINAL — Example hashes

```
hashcat --example-hashes | less
```

Use the official example-hash reference if your version does not support the option. Do not memorize mode numbers without understanding the source.

4. Masks and Rules

Use these only when the lab scenario gives a reason.

KALI TERMINAL — Wordlist + rule

```
hashcat -m [MODE] -a 0 [HASH_FILE] [WORDLIST] -r /usr/share/hashcat/rules/best64.rule
```

KALI TERMINAL — Educational mask example

```
hashcat -m [MODE] -a 3 [HASH_FILE] "?u?l?l?l?d?d?d?d"
```

This example means one uppercase letter, three lowercase letters, and four digits. Do not use a broader mask without a reason.

KALI TERMINAL — Built-in charsets

```
hashcat -h | grep -A 12 "Built-in charsets"
```

5. Sessions and results

KALI TERMINAL — Session

```
hashcat --session=lab1 -m [MODE] -a 0 [HASH_FILE] [WORDLIST]
```

KALI TERMINAL — Restore

```
hashcat --session=lab1 --restore
```

KALI TERMINAL — Status

```
hashcat --status
```

6. Workflow

Step	Decision	Action
1	Identify	Determine hash type and source.
2	Mode	Find the correct -m.
3	Wordlist	Start with -a 0.
4	Rules / Mask	Use only when logically justified.
5	--show	Display cracked results.
6	Validate	Verify only against an authorized account/service.

Prepared as a quick-reference companion to the canonical Arabic guide. Use only in labs or systems you are explicitly authorized to test.

References:

<https://hashcat.net/wiki/doku.php?id=hashcat>

https://hashcat.net/wiki/doku.php?id=example_hashes